Faculty: Computers and Informatics Class Year: 2025-2026 Year: Second Subject Name: Database 1 Subject Code: IS200	Mid Term Exam	Date: 23-11-2025 Time Allowed: 1 hour No. of Pages: 2 Full Mark: 15 Points No of Questions: 8

Name: ..... **Section:** .....

Q1) What are the two main DBMS functions that provide data protection?

Q2) What is the difference between database schema and database state?

Q3) Identify and describe the two types of data independence in DBMS.

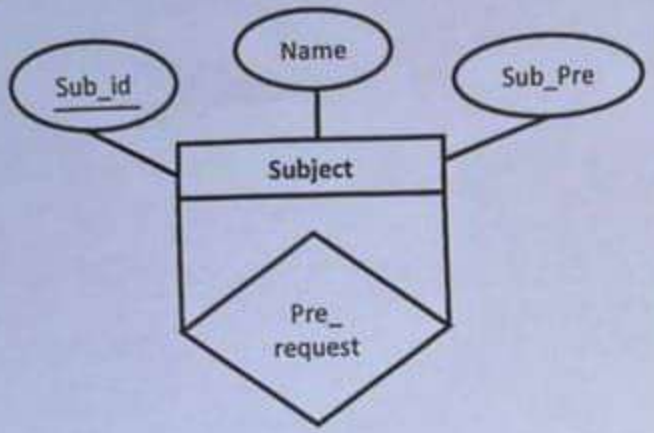
Q4) Draw the complete Enhanced ERD based on the following scenario:

An online marketplace allows **members** to act as either buyers or sellers.

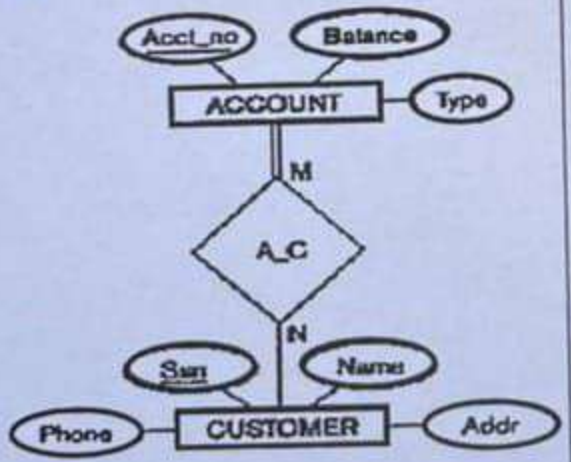
- Each **member** has a unique M_ID, Name composite to Fname and Lname, Email, and one or more Phone Numbers. A member can register as either a **Buyer** or a **Seller**, but not both.
- Each **Buyer** has a Shipping_Address, while each **Seller** has a Bank_Account for receiving payments. **Sellers** list **Items** for sale, each item identified by I_ID, Title, Price, and End_Date.
- Buyers place Orders for items listed on the platform. Each **Order** has an O_ID, O_Amount, and O_Date, and is linked to one buyer and one item.
- A buyer may place multiple orders, and each item may receive several orders.

Answer for EERD:

Q5) Write an SQL statement to create the "Subject" table representing recursive relationship.



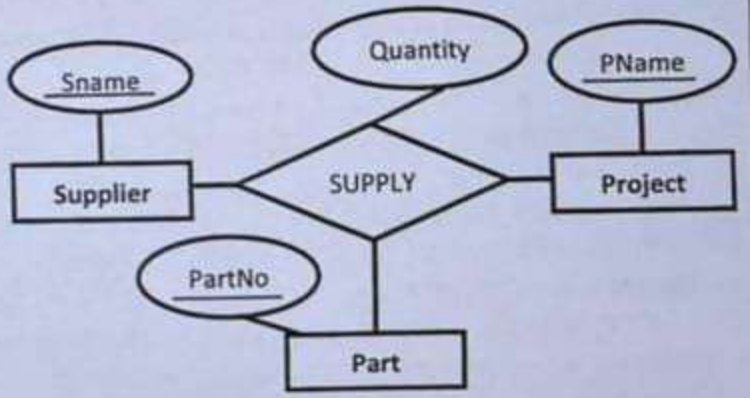
Q6) Write an SQL statement to create the only "A_C" table representing a many-to-many (M:N) relationship.



Q7) Write an SQL statement to create the required tables represented in this relation model.



Q8) Write an SQL statement to create the only "SUPPLY" relation with required constrains of PKs and FKs.



With our best wishes,.....